DfM Analysis Report — DfM Advisor

Model Filename: 0707bdd1-f1e9-4d60-8fa0-63a52640d2c7_Part3.stp
Selected Material: ABS

Metric	Value
Total Face Count	414
Optimal Z Parting Position	16.69 mm
Moldability Score	77.16 / 100
Classification Status	NOT MOLDABLE
Undercuts Count	107
Undercuts Area	1137.94 mm²
Faces requiring geometric splitting	17
Parting Line Length	126.86 mm
Closed Parting Loop	Yes

Engineering Recommendation

The Z = 16.69 mm split minimizes undercut area (1137.9 mm²), reduces faces requiring geometric splitting (17), and produces a highly balanced core/cavity separation. This parting plane yields the highest moldability score (77.2) among all scanned candidates. The physical moldability status is classified as: 'NOT MOLDABLE'.